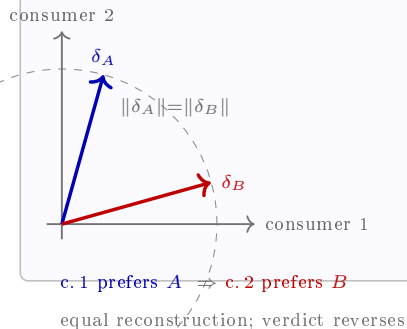


Observation Theory: distortion is what the *consumer* reads, not what the code reconstructs

(a) The flip



(b) The controlling quantity

$$d_{\mathcal{O}} = \text{tr}(P_{\mathcal{C}} \Sigma_{\delta})$$

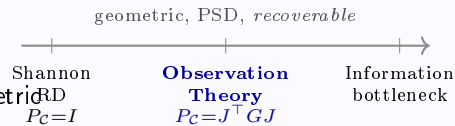
reconstruction error *read through* the consumer

$$P_{\mathcal{C}} = J^{\top} G J = \text{pullback of the consumer's metric}$$

G : output metric J : consumer Jacobian

reconstruction is the isotropic corner $P_{\mathcal{C}} = I$

(c) The identifiable middle term



Recovered blind from consumer queries;
a consumer-relative rate-distortion
function, proven achievable.